Anjuman Institute of Management & Computer Appllications
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award of Degree of BCA

Project Report on

KITCHEN QUEST

Submitted by

Shafaf Ahmed - 20U12718

Under the Guidance of

Prof. Prakash Naik
Prof. Deepa Nayak

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Declaration

We hereby declare that the project report titled **"Kitchen Quest"** was prepared by us during **2022-23**, under the valuable guidance and supervision of **Prof. Prakash Naik** and **Prof. D. N. Noorappanavar (Hasan)** (Lecturers at AIMCA Bhatkal), as a requirement for my Bachelor of Computer Application degree.

we affirm that this project is a manifestation of my our independent work and has not been previously submitted to any other academic institution for the purpose of obtaining any academic degree or recognition.

Signed,
Shafaf Ahmed

Place: Bhatkal
Date: 08/02/2023

Acknowledgement

I would like to express my sincere gratitude to all those who have made this project a success. This project would not have been possible without the guidance, encouragement, and support of several people.

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This project is dedicated to all of them.

Shafaf Ahmed

Abstract

Kitchen Quest is a recipe search website that utilizes HTML5, CSS3, and JavaScript to provide a user-friendly interface for searching and discovering new and delicious recipes. The website uses Edamam API to fetch search results, ensuring up-to-date and accurate recipe information. With Kitchen Quest, users can easily find recipes based on their ingredients, dietary preferences, or cuisines, making meal planning and preparation a breeze. The clean and intuitive design of Kitchen Quest makes it easy for users to find and save their favorite recipes, and the responsive layout ensures a smooth user experience on any device. With Kitchen Quest, the search for the perfect recipe has never been easier.

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Introduction

The purpose of this document is to provide a comprehensive technical specification for the recipe search website, Kitchen Quest. This website is designed to help users find and discover new recipes, easily search for recipes based on ingredients, dietary preferences, or cuisines, and view detailed information about each recipe, including its ingredients, nutritional value, and cooking instructions.

The website has been designed to be intuitive and user-friendly, making it easy for users to navigate and find the information they are looking for. The landing page provides a clear and concise overview of the application's features and functionality.

Website live at :

https://kitchenquest.live

Link to the code(Github Repository) :

https://github.com/

Features

1. **Recipe search:** Allows users to search for recipes by name or ingredients, providing a comprehensive list of relevant results with accompanying images and nutritional information.
2. **Nutritional value:** Displays the nutritional value of each recipe, including calorie count, protein, fat, and carbohydrate content, helping users make informed decisions about the meals they prepare.
3. **Ingredients list:** Provides a list of ingredients for each recipe, making it easy for users to plan and prepare their meals.
4. **User-friendly interface:** Features an intuitive and easy-to-use interface, making it accessible for users of all levels of experience.
5. **Advanced search functionality:** Offers advanced search functionality, allowing users to refine their search results by specifying dietary restrictions, cooking time, and cuisine type.
6. **Responsive design:** The website is designed to be mobile-friendly and responsive, allowing users to access it from a variety of devices and platforms.

System Study

The aim of the System Study phase is to determine whether the proposed system can be developed and implemented within the given resources and constraints. The System Study of the Kitchen Quest application includes the following stages:

Proposed System With Objectives:

Kitchen Quest is a recipe search website that utilizes HTML5, CSS3, and JavaScript to provide a user-friendly interface for searching and discovering new and delicious recipes. The website uses Edamam API to fetch search results, ensuring up-to-date and accurate recipe information. With Kitchen Quest, users can easily find recipes based on their ingredients, dietary preferences, or cuisines, making meal planning and preparation a breeze. The clean and intuitive design of Kitchen Quest makes it easy for users to find and save their favorite recipes, and the responsive layout ensures a smooth user experience on any device. With Kitchen Quest, the search for the perfect recipe has never been easier.

Problems in Existing system :

Current recipe search websites often lack a clean and intuitive design, making it difficult for users to easily find and save their favorite recipes. Additionally, many websites do not provide detailed information about each recipe, such as its ingredients and nutritional value, making meal planning and preparation more challenging.

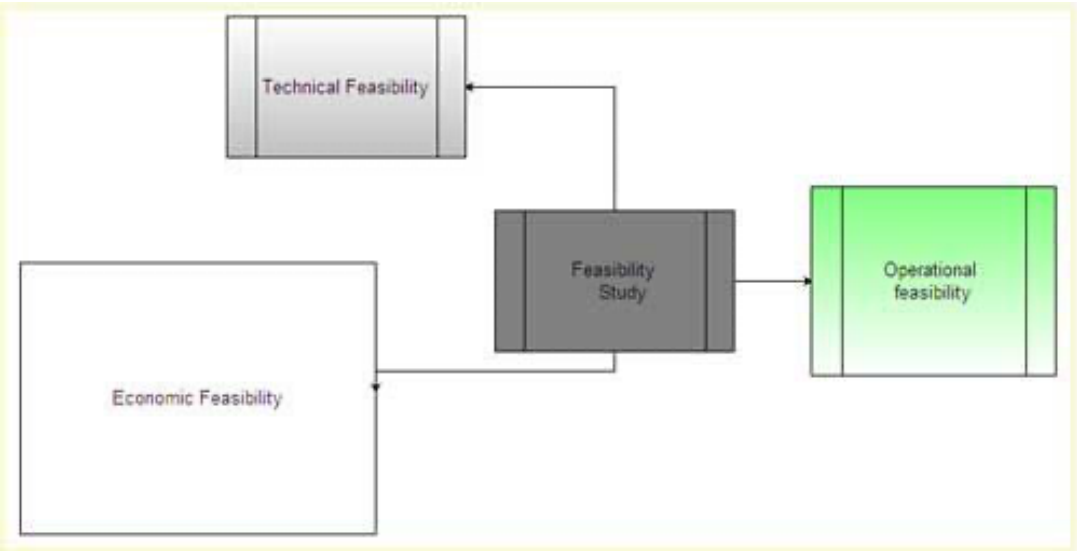
Objectives of the system :

The main objectives of the system are to provide a user-friendly interface for searching and discovering new recipes, to provide accurate and up-to-date information about each recipe, and to make meal planning and preparation easier for users.

Feasibility Studies

Conceivable plausibility ponders are conceivable to bring together all the conceivable work prerequisites to test all the fundamental assets and dangers present in the earth for the improvement of our frameworks. The likelihood for arranging progressively sorted out data will be isolated into various segments of concentrate with the goal that the reason can be reinforced and the correct methodology for the data association can be masterminded which will assist us with making choices and can be talked about.

The achievability contemplate is significant for safeguarding of possibility and organisations will almost certainly deal with the task productively for the standard contemplations that ought to be talked about for the parametric structure.



Technical Feasibility

The technical feasibility of the system is determined by the availability of the necessary technology and infrastructure, including the Edamam API and web development technologies such as HTML5, CSS3, and JavaScript.
Hosting and Domain Requirements: Determine the hosting and domain requirements for the website, including disk space, bandwidth, and security.
Browser Compatibility: Ensure that the website is compatible with the most commonly used browsers, such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari.

Operational Feasibility

Time for Updates and Maintenance: Evaluate the amount of time required for regular updates and maintenance, including fixing bugs, adding new features, and updating portfolio projects and games.

Ease of Updating Portfolio Projects and Games: Determine the ease of updating the portfolio projects and games, including adding new projects, updating descriptions and images, and removing old projects.
Ease of Adding New Features: Evaluate the ease of adding new features to the website, such as new pages, functionality, or design elements.

Economic Feasibility

Cost of Tools and Technologies: Consider the cost of purchasing or subscribing to the tools and technologies needed to build the website, including any software licenses or development tools.

Hosting and Domain Expenses: Determine the cost of purchasing a domain and hosting the website, including any recurring fees and additional services.
Maintenance Costs: Consider the cost of maintaining and updating the website, including fixing bugs, adding new features, and updating portfolio projects and games.

System Design

Systems design is the process of defining the architecture, interfaces, and data for a system that satisfies requirements. System design should satisfy specific needs of a business or organisation through a coherent, well-running system.

After you have your requirements, you examine and transform them into a physical system design that addresses the customers' needs. The design activity will vary depending on if you go for custom development, commercial solutions, or a combination of both.

Systems design requires a systematic approach to building and engineering a system. A good system design requires engineers to think about everything in an infrastructure, from the hardware/software, down to the data and how it's stored.

System design includes the following design methods:

- Architectural design: describes the views, models, behaviour, and infrastructure of a system.
- Logical design: represents the data flow and inputs/outputs of a system.
- Physical design: includes how users can add information, how a system represents information to users, and how data is modelled/stored.

Sytem Configurations

Hardware

Pentium processor : 512 MHZ

RAM Capacity : 2GB

Hard Disk : 40GB

CD-ROM Drive : 32 HZ

Software

Operating system : Windows / Linux / android / mac

Platform used : Front-end development

IDE : Visual Studio code 2021

Language Used : HTML, CSS , JAVASCRIPT

Tools : Figma

Browser : Google Chrome: 89

Mozilla Firefox: 87

Microsoft Edge 91

Safari: 14.2

Languages and Tools used

HTML

HTML (Hypertext Markup Language) is a standard markup language used for creating web pages. It provides the structure for displaying content on the internet, including text, images, videos, and links.

In the Kitchen Quest project, HTML is used as the main structure for building the user interface of the web application. It provides the structure for the different components of the website, such as the header, navigation menu, main content area, and footer.

HTML5, the latest version of HTML, has several new features that make it easier to create more interactive and engaging web pages. Some of the key features of HTML5 include:

1. Semantic Elements: HTML5 introduces new semantic elements, such as header, footer, article, section, and nav, which make it easier to structure content and provide context to search engines.
2. Form Validation: HTML5 includes new form controls and validation features that make it easier to create forms and validate user input.
3. Video and Audio: HTML5 allows for native support for video and audio, making it easier to add media content to a web page without the need for additional plugins.
4. Canvas: The HTML5 canvas element provides a way to draw graphics on the web page using JavaScript.
5. Geolocation: HTML5 allows websites to access the user's location, which is useful for location-based applications.

CSS

CSS (Cascading Style Sheets) is an essential part of the Kitchen Quest project. CSS is used to define the look and feel of the application, including its layout, colors, fonts, and overall visual appearance.
CSS has several features that make it a powerful tool for designing websites and applications. Some of the key features of CSS include:

1. Separation of presentation and content: CSS allows developers to separate the visual presentation of a website or application from its content, making it easier to maintain and update the appearance of the application.
2. Reusability: CSS styles can be reused across multiple pages, reducing the time and effort required to make changes to the overall appearance of the application.
3. Consistency: CSS enables developers to create consistent visual styles across all pages of a website or application, making it easier for users to navigate and interact with the application.

JavaScript

JavaScript is an essential part of the Kitchen Quest project. JavaScript is a programming language that is used to add interactivity and dynamic functionality to websites and applications.
In the Kitchen Quest project, JavaScript is used to bring the application to life, enabling users to interact with the application and access its features in real-time. JavaScript provides several key features that are critical to the success of the Kitchen Quest project, including:

1. Dynamic Functionality: JavaScript enables developers to create dynamic and interactive features, such as animations, real-time updates, and user-triggered events.
2. Interactivity: JavaScript provides tools for creating an interactive user interface, allowing users to interact with the application and access its features in real-time.

3. Validation: JavaScript can be used to validate user inputs, ensuring that the data entered by users is correct and complete before it is submitted.
4. Cross-browser Compatibility: JavaScript is supported by all major web browsers, making it an ideal choice for developing cross-browser compatible applications.

Git & Github

Git/Github is a version control system and a web-based platform for hosting and sharing code. In the context of the Kitchen Quest project, Git/Github can be used to manage and store the code for the application. The use of Git/Github allows for collaboration between multiple developers, tracking changes to the code, and the ability to revert back to previous versions of the code if necessary.
Some of the key features of Git/Github include:

1. Version Control: Git allows developers to keep track of all changes made to the code over time and provides a way to revert back to previous versions if needed.
2. Branching and Merging: With Git, developers can create separate branches for different features or bug fixes, and then merge these branches back into the main codebase when the work is complete.
3. Collaboration: Github provides a web-based platform for multiple developers to work on the same codebase, track changes, and communicate with each other.
4. Pull Requests: Github provides a way for developers to submit changes to the code for review and approval by other team members before merging into the main codebase.
5. History and Auditing: Git/Github allows developers to view a complete history of all changes made to the code, making it easier to track down bugs and understand how the code evolved over time.

Edamam API

The Edamam API is a crucial component of the Kitchen Quest project as it provides the recipe search functionality of the website. The API allows the website to retrieve recipe information, such as ingredients, nutritional information, and cooking instructions, from a database of over 1.5 million recipes.

Some of the key features of the Edamam API include:

1. Recipe Database: The Edamam API provides access to a large and diverse database of over 1.5 million recipes, making it a valuable resource for recipe search and discovery.
2. Nutritional Information: The API provides detailed nutritional information for each recipe, including calorie count, protein, fat, and carbohydrate content, among other things. This information is especially useful for users who are following specific dietary restrictions or trying to maintain a healthy diet.
3. Search Functionality: The API provides advanced search functionality, allowing users to search for recipes by name, ingredients, and other criteria. This makes it easy for users to find exactly what they are looking for and discover new and exciting recipes to try.
4. API Access: The Edamam API is easy to integrate into the Kitchen Quest website, allowing for seamless and reliable access to the recipe database and search functionality.

Overall, the Edamam API is a critical component of the Kitchen Quest project, providing the recipe search and discovery functionality that is essential to the success of the website. With its vast recipe database, advanced search functionality, and easy API access, the Edamam API provides a valuable resource for users seeking new and exciting recipes to try.

BoxIcons - Icons

Boxicons is a free, open-source icon set that can be used in web design and development projects. The icon set includes a wide range of icons for various applications, such as navigation, user interface elements, social media, and more. The icons are designed to be simple, clean, and consistent, making them easy to integrate into any web design.

Visual Studio Code

Visual Studio Code (VS Code) is a source code editor developed by Microsoft. It is used as an integrated development environment (IDE) for developing software applications. In the context of the "Kitchen Quest" project, VS Code is used as the primary code editor for developing the website.

Some of the features of VS Code that make it an ideal choice for this project include:

1. IntelliSense: VS Code provides intelligent code completion and suggestions that are powered by AI algorithms. This helps the developer write code faster and with fewer errors.
2. Debugging: VS Code includes a powerful debugger that helps developers identify and fix bugs in their code.
3. Git Integration: As Git is a popular version control system, VS Code integrates seamlessly with Git. This allows developers to easily manage their codebase and track changes.
4. Extensions: VS Code has a vast library of extensions available that can add new functionalities to the code editor. For example, the "Bootstrap 4, Font awesome 5" extension can be used to add the Bootstrap framework to the project.
5. User-friendly interface: The VS Code interface is clean, easy to navigate and customize, making it a great choice for developers of all skill levels.

Database Design

The proposed system does not require any database as the data is fetched using edamam api.

Table Design

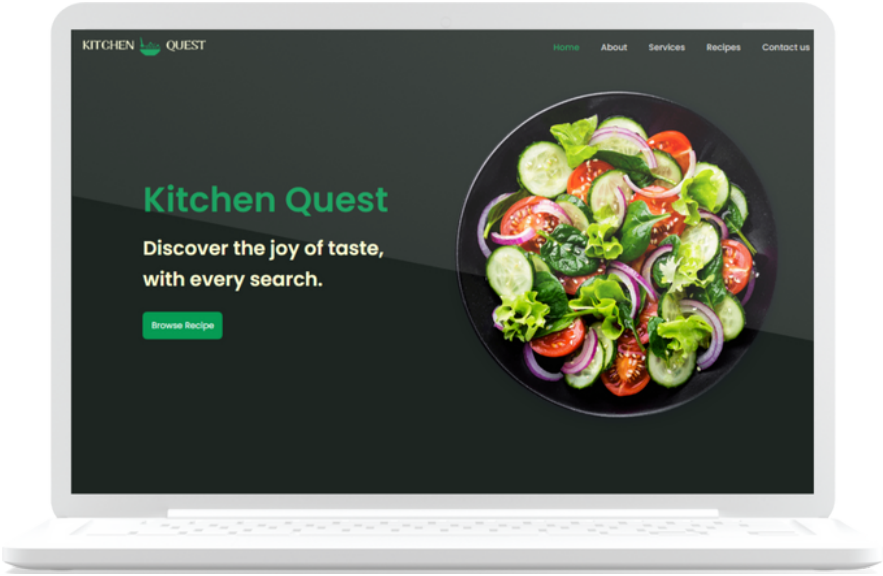
No tables are required for the proposed system as the browser local storage provides the necessary storage mechanism.

Screen Design

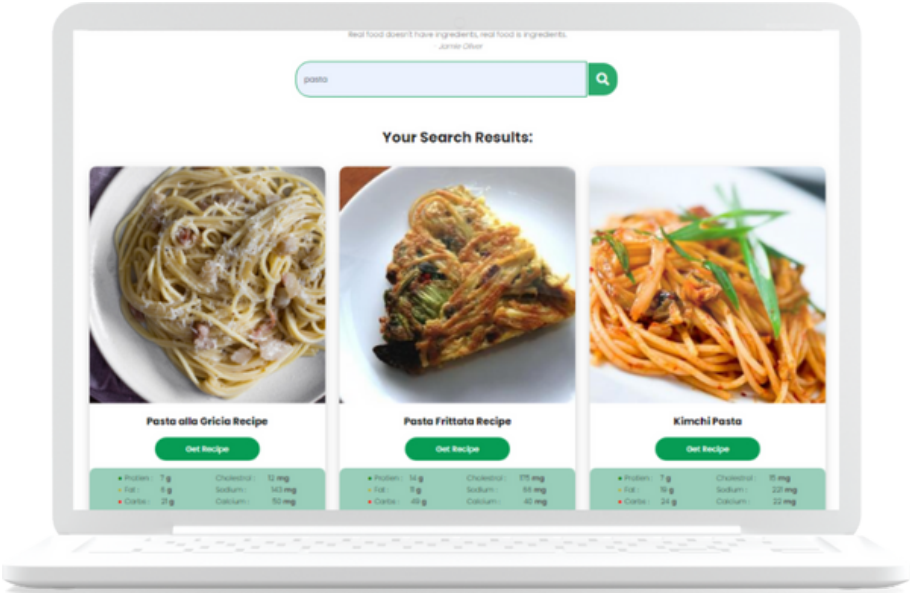
The proposed system consists of three pages: the landing page , the recipe search page and the contact page. The landing page provides users with a brief overview of the website and its features, the recipe page allows users to search for recipes. and the contact page provides a form to contact.

System UI

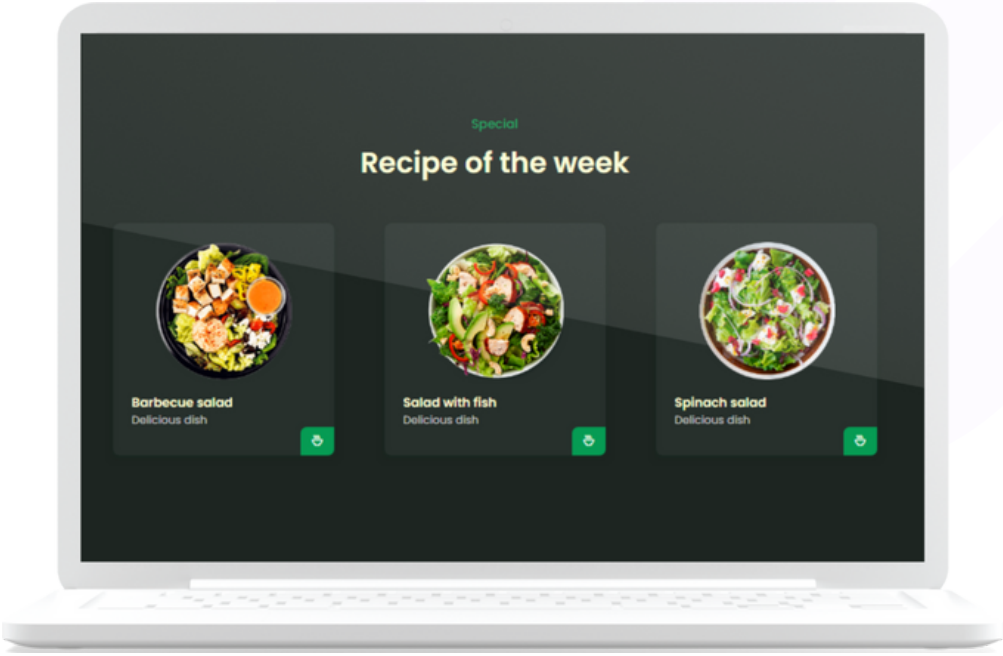
Landing page



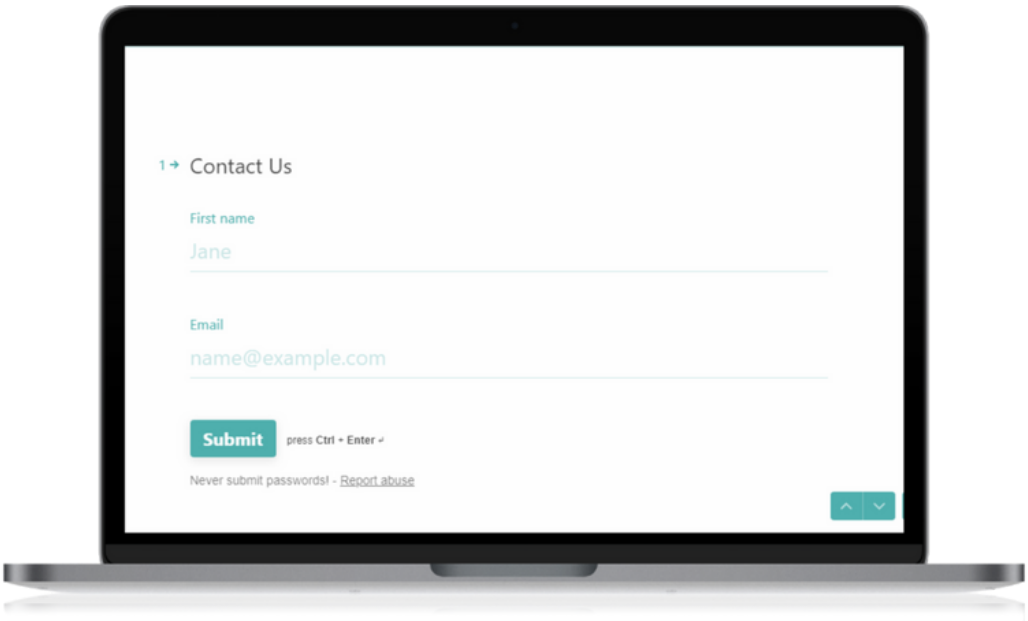
Recipe page



Recipes of the week



Contact form



System Testing

Testing steps

The following steps will be followed in testing the proposed system:

1. **Unit testing :** testing individual components of the system to ensure they work as expected.
2. **Integration testing :** testing the interaction between different components of the system.
3. **User acceptance testing** : testing the system with real users to ensure it meets their expectations and requirements.

Stages in Testing Process

The testing process for the proposed system consists of the following stages:

1. **Planning and preparation:** defining the testing objectives and creating a testing plan.
2. **Test case design:** designing the test cases to be used in the testing process.
3. **Test execution:** executing the test cases and recording the results.
4. **Test result analysis**: analyzing the results of the tests to identify any errors or bugs.
5. **Final evaluation:** evaluating the system based on the results of the tests to determine if it meets the required specifications.

Future Enhancements

The Kitchen Quest recipe search website has a lot of potential for future development and expansion. Some potential areas for future growth include:

1. Recipe Submission: Allowing users to submit their own recipes, along with photos and instructions, would help to build a community of food enthusiasts and expand the variety of recipes available on the website.
2. Meal Planning: Integrating meal planning functionality, such as the ability to create shopping lists and plan out meals for a week, would make the website a comprehensive resource for home cooks.
3. Recipe Recommendations: Utilizing machine learning algorithms to make personalized recipe recommendations based on a user's search history and favorites would make the recipe search experience even more personalized and efficient.
4. Social Integration: Allowing users to share their favorite recipes and meals on social media platforms would help to expand the reach of the website and build a community of food enthusiasts.
5. Mobile Application: Developing a mobile application for the website would make it easier for users to access the recipes and meal planning tools on-the-go.
6. Nutritional Tracking: Adding the ability for users to track their daily nutrient intake and set goals for themselves would make the website a comprehensive resource for healthy eating and meal planning.
7. User Accounts: Implementing user accounts, with the ability to save favorite recipes, meal plans, and track progress, would enhance the user experience and encourage repeat visits to the website.

Conclusion

The Kitchen Quest recipe search website was developed with the goal of providing users with an easy-to-use platform for searching for and finding delicious and nutritious recipes. The website was built using HTML5, CSS3, and Javascript and utilizes the Edamam API to fetch search results. With its user-friendly interface and advanced search functionality, Kitchen Quest is a valuable resource for home cooks of all levels of experience.

The website has potential for future growth and development, including the integration of meal planning, social media sharing, and nutritional tracking. The website has the potential to become a comprehensive resource for home cooks, providing all the tools and information they need to plan and prepare delicious and nutritious meals.

Bibliography

1. HTML , CSS and JavaScript :

100DaysOfWebDevelopment by Max - Udemy

https://www.udemy.com/course/100-days-of-code-web-development-bootcamp/

2. HTML Local Storage :

W3Schools

https://www.w3schools.com/html/html5_webstorage.asp

3. Assistance

ChatGPT

https://chat.openai.com/chat

4. API

Edamam

https://chat.openai.com/chat

Thank you

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